

Data Preprocessing? — Major Tasks

- ❑ **Data cleaning**

- ❑ Handle missing data, smooth noisy data, identify or remove outliers, and resolve inconsistencies

- ❑ **Data integration**

- ❑ Integration of multiple databases, data cubes, or files

- ❑ **Data reduction**

- ❑ Dimensionality reduction
 - ❑ Numerosity reduction
 - ❑ Data compression

- ❑ **Data transformation and data discretization**

- ❑ Normalization
 - ❑ Concept hierarchy generation

Why Preprocess the Data? — Data Quality Issues

- ❑ Measures for data quality: A multidimensional view
 - ❑ Accuracy: correct or wrong, accurate or not
 - ❑ Completeness: not recorded, unavailable, ...
 - ❑ Consistency: some modified but some not, dangling, ...
 - ❑ Timeliness: timely update?
 - ❑ Believability: how trustable the data are correct?
 - ❑ Interpretability: how easily the data can be understood?

Data Cleaning

- ❑ Data in the Real World Is Dirty: Lots of potentially incorrect data, e.g., instrument faulty, human or computer error, and transmission error
- ❑ Incomplete: lacking attribute values, lacking certain attributes of interest, or containing only aggregate data
 - ❑ e.g., *Occupation* = " " (missing data)
- ❑ Noisy: containing noise, errors, or outliers
 - ❑ e.g., *Salary* = "-10" (an error)
- ❑ Inconsistent: containing discrepancies in codes or names, e.g.,
 - ❑ *Age* = "42", *Birthday* = "03/07/2010"
 - ❑ Was rating "1, 2, 3", now rating "A, B, C"
 - ❑ discrepancy between duplicate records
- ❑ Intentional (e.g., *disguised missing* data)
 - ❑ Jan. 1 as everyone's birthday?

Incomplete (Missing) Data

- ❑ Data is not always available
 - ❑ E.g., many tuples have no recorded value for several attributes, such as customer income in sales data
- ❑ Missing data may be due to
 - ❑ Equipment malfunction
 - ❑ Inconsistent with other recorded data and thus deleted
 - ❑ Data were not entered due to misunderstanding
 - ❑ Certain data may not be considered important at the time of entry
 - ❑ Did not register history or changes of the data
- ❑ Missing data may need to be inferred

Filling-up the missing values

	Outlook	Temperature	Humidity	Windy	PlayGolf
0	sunny	85	85	FALSE	no
1	rainy	80	90	TRUE	no
2	overcast	83	86	FALSE	yes
3	rainy	70	96	FALSE	yes
4	rainy	72	80	FALSE	yes
5	rainy	65	70	TRUE	no
6	overcast	64	65	TRUE	yes
7	sunny	72	95	FALSE	no
8	sunny	69	70	FALSE	yes
9	rainy	75	80	FALSE	yes
10	sunny	75	70	TRUE	yes
11	overcast	72	80	TRUE	yes
12	overcast	81	75	FALSE	yes
13	rainy	71	91	TRUE	no

How to Handle Missing Data?

- ❑ Ignore the tuple: usually done when class label is missing (when doing classification)—not effective when the % of missing values per attribute varies considerably
- ❑ Fill in the missing value manually: tedious + infeasible?
- ❑ Fill in it automatically with
 - ❑ a global constant : e.g., “unknown”, a new class?!
 - ❑ the attribute mean
 - ❑ the attribute mean for all samples belonging to the same class: smarter
 - ❑ **the most probable value: inference-based such as Bayesian formula or decision tree**

Noisy Data

- ❑ **Noise:** random error or variance in a measured variable
- ❑ **Incorrect attribute values** may be due to
 - ❑ Faulty data collection instruments
 - ❑ Data entry problems
 - ❑ Data transmission problems
 - ❑ Technology limitation
 - ❑ Inconsistency in naming convention
- ❑ **Other data problems**
 - ❑ Duplicate records
 - ❑ Incomplete data
 - ❑ Inconsistent data

How to Handle Noisy Data?

- ❑ Binning
 - ❑ First sort data and partition into (equal-frequency) bins
 - ❑ Then one can **smooth by bin means, smooth by bin median, smooth by bin boundaries**, etc.
- ❑ Regression
 - ❑ Smooth by fitting the data into regression functions
- ❑ Clustering
 - ❑ Detect and remove outliers
- ❑ Semi-supervised: Combined computer and human inspection
 - ❑ Detect suspicious values and check by human (e.g., deal with possible outliers)

Binning

- **Equal-depth** partitioning:
 - It divides the data such that each bin contains approximately same number of samples
- **Equal-width** partitioning:
 - It divides the range into N intervals of equal size: uniform grid
 - if A and B are the lowest and highest values of the attribute, the width of intervals will be: $W = (B-A)/N$
 - Where 'N' is the number of bins
 - Skewed data is not handled well.

Equi-depth Binning

* Data Given : 4, 8, 9, 15, 21, 21, 24, 25, 26, 28, 29, 34

* Partition into (equi-depth) bins:

- Bin 1: 4, 8, 9, 15
- Bin 2: 21, 21, 24, 25
- Bin 3: 26, 28, 29, 34

* Smoothing by bin means:

- Bin 1: 9, 9, 9, 9
- Bin 2: 23, 23, 23, 23
- Bin 3: 29, 29, 29, 29

* Smoothing by bin boundaries:

- Bin 1: 4, 4, 4, 15
- Bin 2: 21, 21, 25, 25
- Bin 3: 26, 26, 26, 34

Smoothing by bin median:

- Bin 1: 8.5, 8.5, 8.5, 8.5
- Bin 2: 22.5, 22.5, 22.5, 22.5
- Bin 3: 28.5, 28.5, 28.5, 28.5

Equi-Width Binning

- Bins have equal width with a range of each bin are defined as $[\min + w]$, $[\min + 2w]$ $[\min + nw]$, where $w = (\max - \min) / N$
- Given Data Set ; 5, 10, 11, 13, 15, 35, 50 ,55, 72, 92, 204, 215
- Width = $(215 - 5) / 3 = 70$

Bin 1 : $5 + 70 = 75$: 5,10,11,13,15,35,50,55,72

Bin 2 : $5 + 140 = 145$: 92

Bin 3 : $5 + 210 = 215$: 204,215

Do the Following...

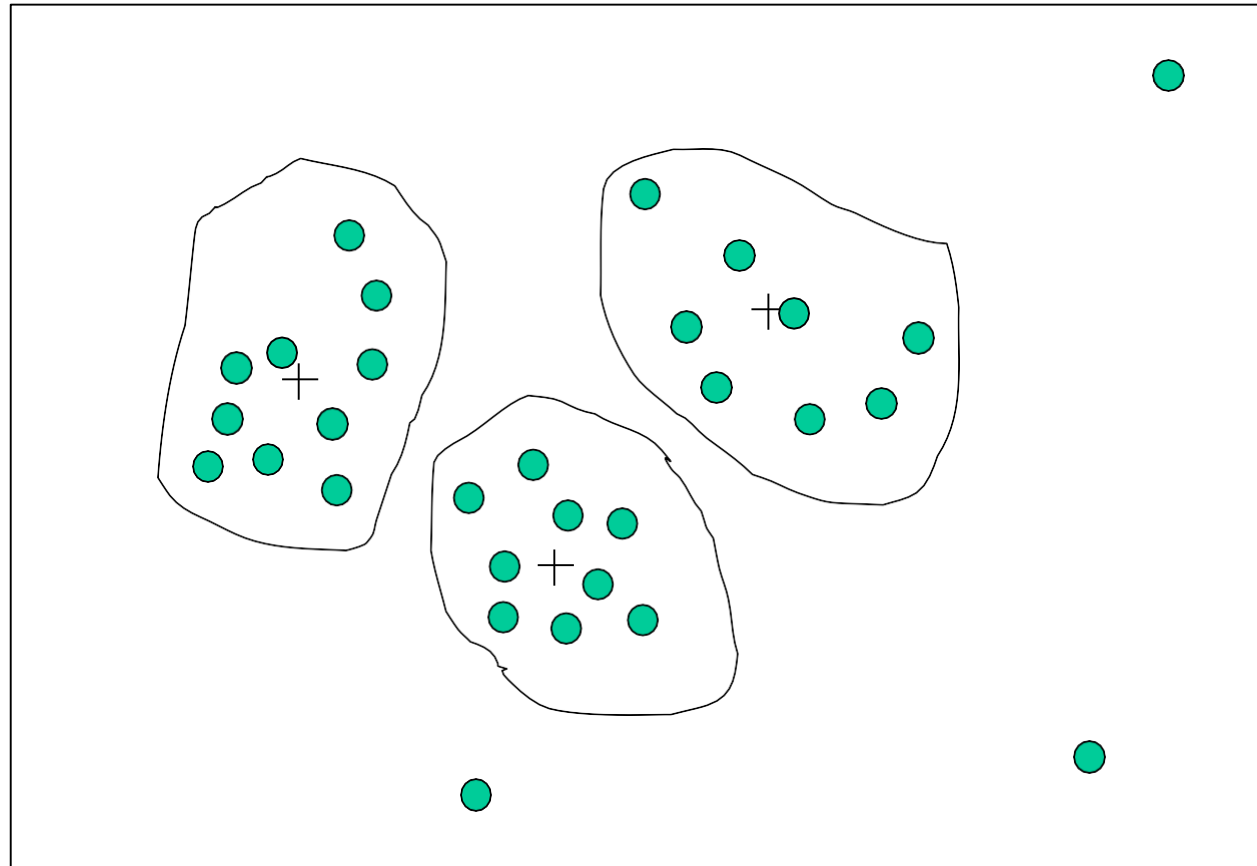
Data:

11,13,13,15,15,16,19,20,20,20,21,
21,22,23,24,30,40,45,45,45,45,71,
72,73,75

Do the Following based on Equi-Depth, and Equi-Width Binning methods by choosing 4 Bins

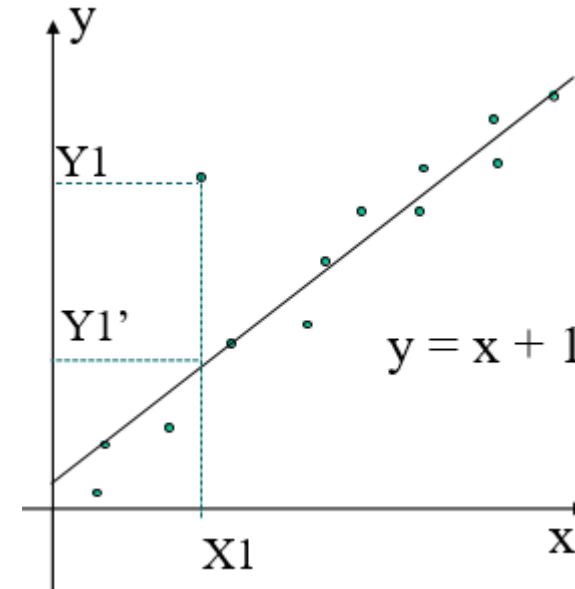
- a) Smoothing by bin mean
- b) Smoothing by bin median
- c) Smoothing by bin boundaries

Cluster Analysis



Regression Analysis

- Regression analysis: A collective name for techniques for the modeling and analysis of numerical data consisting of values of a **dependent variable** (also called **response variable** or *measurement*) and of one or more **independent variables** (aka. **explanatory variables** or **predictors**)
- The parameters are estimated so as to give a "**best fit**" of the data
- Most commonly the best fit is evaluated by using the **least squares method**, but other criteria have also been used



- Used for prediction (including forecasting of time-series data), inference, hypothesis testing, and modeling of causal relationships

Data Cleaning as a Process

❑ Data discrepancy detection

- ❑ Use metadata (e.g., domain, range, dependency, distribution)
- ❑ Check field overloading
- ❑ Check uniqueness rule, consecutive rule and null rule
- ❑ Use commercial tools
 - ❑ Data scrubbing: use simple domain knowledge (e.g., postal code, spell-check) to detect errors and make corrections
 - ❑ Data auditing: by analyzing data to discover rules and relationship to detect violators (e.g., correlation and clustering to find outliers)

❑ Data migration and integration

- ❑ Data migration tools: allow transformations to be specified
 - ❑ ETL (Extraction/Transformation/Loading) tools: allow users to specify transformations through a graphical user interface
- ## ❑ Integration of the two processes
- ❑ Iterative and interactive (e.g., Potter's Wheels)

Data Integration

- ❑ Data integration
 - ❑ Combining data from multiple sources into a coherent store
- ❑ Why data integration?
 - ❑ Help reduce/avoid noise
 - ❑ Get a more complete picture
 - ❑ Improve mining speed and quality
- ❑ **Schema integration:**
 - ❑ e.g., $A.cust-id \equiv B.cust-\#$
 - ❑ Integrate metadata from different sources
- ❑ **Entity identification:**
 - ❑ Identify real world entities from multiple data sources, e.g., Bill Clinton = William Clinton

Handling Noise in Data Integration

- ❑ Detecting data value conflicts
 - ❑ For the same real world entity, attribute values from different sources are different
 - ❑ Possible reasons: no reason, different representations, different scales, e.g., metric vs. British units
- ❑ Resolving conflict information
 - ❑ Take the mean/median/mode/max/min
 - ❑ Take the most recent
 - ❑ Truth finding: consider the source quality
- ❑ Data cleaning + data integration

Handling Redundancy in Data Integration

- ❑ Redundant data occur often when integration of multiple databases
 - ❑ *Object identification*: The same attribute or object may have different names in different databases
 - ❑ *Derivable data*: One attribute may be a “derived” attribute in another table, e.g., annual revenue
- ❑ What’s the problem?
 - ❑ $Y = 2X \rightarrow Y = X_1 + X_2 \quad Y = 3X_1 - X_2 \quad Y = -1291X_1 + 1293X_2$
- ❑ Redundant attributes may be detected by correlation analysis and covariance analysis

Data Transformation

- ☐ Normalization
- ☐ Discretization
- ☐ Data Compression
- ☐ Sampling

Data Transformation

- ❑ A function that maps the entire set of values of a given attribute to a new set of replacement values s.t. each old value can be identified with one of the new values
- ❑ Methods
 - ❑ Smoothing: Remove noise from data
 - ❑ Attribute/feature construction
 - ❑ New attributes constructed from the given ones
 - ❑ Aggregation: Summarization, data cube construction
 - ❑ Normalization: Scaled to fall within a smaller, specified range
 - ❑ min-max normalization
 - ❑ z-score normalization
 - ❑ normalization by decimal scaling
 - ❑ Discretization: Concept hierarchy climbing

Normalization

- **Min-max normalization:** to $[\text{new_min}_A, \text{new_max}_A]$

$$v' = \frac{v - \text{min}_A}{\text{max}_A - \text{min}_A} (\text{new_max}_A - \text{new_min}_A) + \text{new_min}_A$$

- Ex. Let income range \$12,000 to \$98,000 normalized to [0.0, 1.0]

□ Then \$73,000 is mapped to $\frac{73,600 - 12,000}{98,000 - 12,000} (1.0 - 0) + 0 = 0.716$

- **Z-score normalization** (μ : mean, σ : standard deviation):

$$v' = \frac{v - \mu_A}{\sigma_A}$$

Z-score: The distance between the raw score and the population mean in the unit of the standard deviation

- Ex. Let $\mu = 54,000$, $\sigma = 16,000$. Then $\frac{73,600 - 54,000}{16,000} = 1.225$

- **Normalization by decimal scaling**

$$v' = \frac{v}{10^j} \quad \text{Where } j \text{ is the smallest integer such that } \text{Max}(|v'|) < 1$$

Normalize the Following Data using Min-Max, Z-Score, and Decimal Scaling Methods

Variable
33
21
7
53
29
42
12
19
22
36

Discretization

- ❑ Three types of attributes
 - ❑ Nominal—values from an unordered set, e.g., color, profession
 - ❑ Ordinal—values from an ordered set, e.g., military or academic rank
 - ❑ Numeric—real numbers, e.g., integer or real numbers
- ❑ Discretization: Divide the range of a continuous attribute into intervals
 - ❑ Interval labels can then be used to replace actual data values
 - ❑ Reduce data size by discretization
 - ❑ Supervised vs. unsupervised
 - ❑ Split (top-down) vs. merge (bottom-up)
 - ❑ Discretization can be performed recursively on an attribute
 - ❑ Prepare for further analysis, e.g., classification

Data Discretization Methods

- ❑ Binning
 - ❑ Top-down split, unsupervised
- ❑ Histogram analysis
 - ❑ Top-down split, unsupervised
- ❑ Clustering analysis
 - ❑ Unsupervised, top-down split or bottom-up merge
- ❑ Decision-tree analysis
 - ❑ Supervised, top-down split
- ❑ Correlation (e.g., χ^2) analysis
 - ❑ Unsupervised, bottom-up merge
- ❑ Note: All the methods can be applied recursively

Simple Discretization: Binning

- ❑ **Equal-width** (distance) partitioning
 - ❑ Divides the range into N intervals of equal size: uniform grid
 - ❑ if A and B are the lowest and highest values of the attribute, the width of intervals will be: $W = (B - A)/N$.
 - ❑ The most straightforward, but outliers may dominate presentation
 - ❑ Skewed data is not handled well
- ❑ **Equal-depth** (frequency) partitioning
 - ❑ Divides the range into N intervals, each containing approximately same number of samples
 - ❑ Good data scaling
 - ❑ Managing categorical attributes can be tricky

Example: Binning Methods for Data Smoothing

❑ Sorted data for price (in dollars): 4, 8, 9, 15, 21, 21, 24, 25, 26, 28, 29, 34

* Partition into equal-frequency (**equal-depth**) bins:

- Bin 1: 4, 8, 9, 15
- Bin 2: 21, 21, 24, 25
- Bin 3: 26, 28, 29, 34

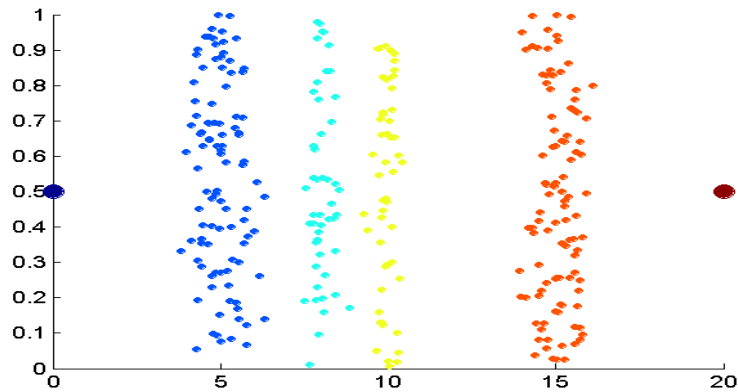
* Smoothing by **bin means**:

- Bin 1: 9, 9, 9, 9
- Bin 2: 23, 23, 23, 23
- Bin 3: 29, 29, 29, 29

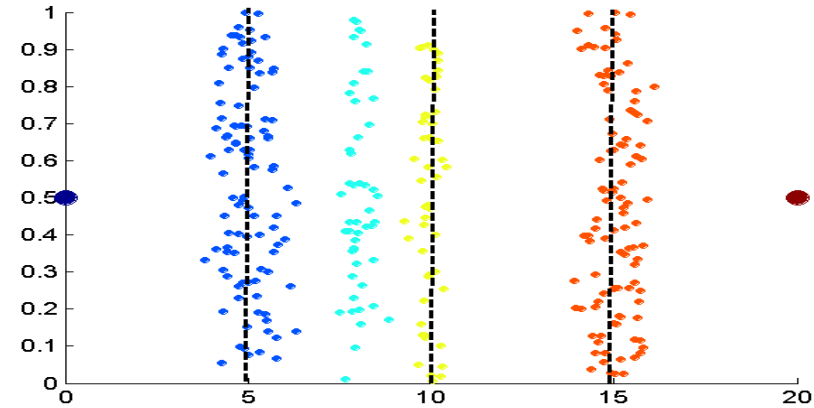
* Smoothing by **bin boundaries**:

- Bin 1: 4, 4, 4, 15
- Bin 2: 21, 21, 25, 25
- Bin 3: 26, 26, 26, 34

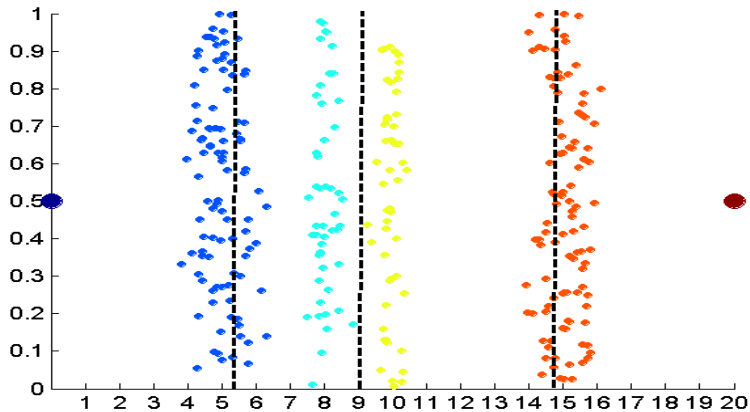
Discretization Without Supervision: Binning vs. Clustering



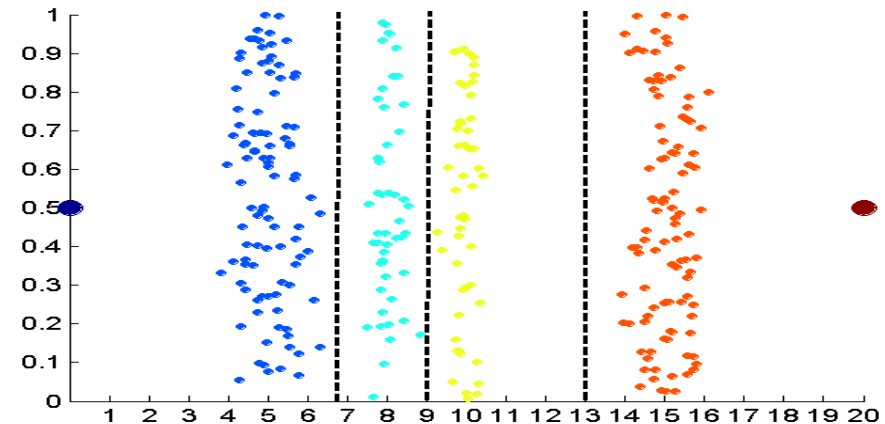
Data



Equal width (distance) binning



Equal depth (frequency) (binning)



K-means clustering leads to better results

Discretization by Classification & Correlation Analysis

- ❑ Classification (e.g., decision tree analysis)
 - ❑ Supervised: Given class labels, e.g., cancerous vs. benign
 - ❑ Using *entropy* to determine split point (discretization point)
 - ❑ Top-down, recursive split
 - ❑ Details to be covered in Chapter “Classification”
- ❑ Correlation analysis (e.g., Chi-merge: χ^2 -based discretization)
 - ❑ Supervised: use class information
 - ❑ Bottom-up merge: Find the best neighboring intervals (those having similar distributions of classes, i.e., low χ^2 values) to merge
 - ❑ Merge performed recursively, until a predefined stopping condition

Concept Hierarchy Generation

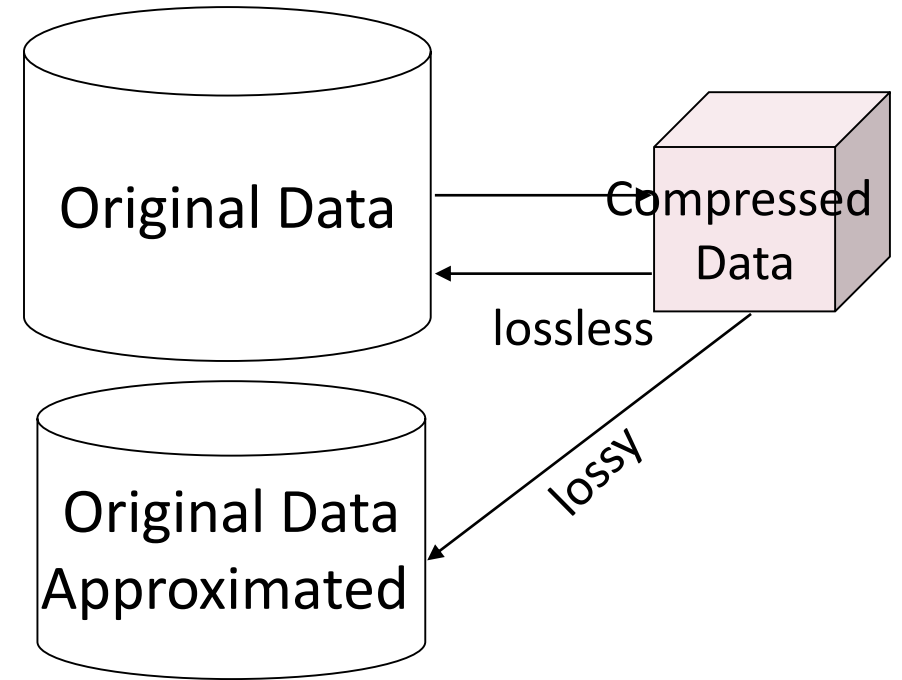
- ❑ **Concept hierarchy** organizes concepts (i.e., attribute values) hierarchically and is usually associated with each dimension in a data warehouse
- ❑ Concept hierarchies facilitate drilling and rolling in data warehouses to view data in multiple granularity
- ❑ Concept hierarchy formation: Recursively reduce the data by collecting and replacing low level concepts (such as numeric values for *age*) by higher level concepts (such as *youth, adult, or senior*)
- ❑ Concept hierarchies can be explicitly specified by domain experts and/or data warehouse designers
- ❑ Concept hierarchy can be automatically formed for both numeric and nominal data—For numeric data, use discretization methods shown

Concept Hierarchy Generation for Nominal Data

- ❑ Specification of a partial/total ordering of attributes explicitly at the schema level by users or experts
 - ❑ *street < city < state < country*
- ❑ Specification of a hierarchy for a set of values by explicit data grouping
 - ❑ {Urbana, Champaign, Chicago} < Illinois
- ❑ Specification of only a partial set of attributes
 - ❑ E.g., only *street < city*, not others
- ❑ Automatic generation of hierarchies (or attribute levels) by the analysis of the number of distinct values
 - ❑ E.g., for a set of attributes: {*street, city, state, country*}

Data Compression

- ❑ String compression
 - ❑ There are extensive theories and well-tuned algorithms
 - ❑ Typically lossless, but only limited manipulation is possible without expansion
- ❑ Audio/video compression
 - ❑ Typically lossy compression, with progressive refinement
 - ❑ Sometimes small fragments of signal can be reconstructed without reconstructing the whole
- ❑ Time sequence is not audio
 - ❑ Typically short and vary slowly with time
- ❑ Data reduction and dimensionality reduction may also be considered as forms of data compression



Lossy vs. lossless compression

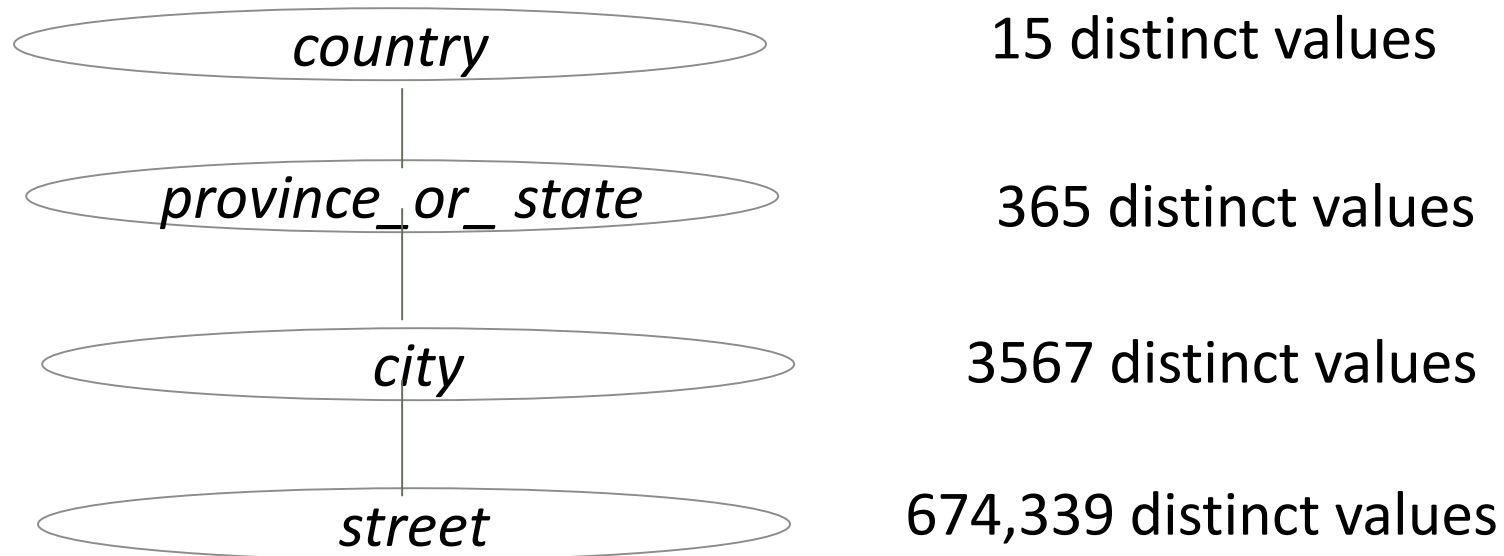
Data Cube Aggregation

- The lowest level of a data cube (base cuboid)
 - The aggregated data for an **individual entity of interest**
 - E.g., a customer in a phone calling data warehouse
- Multiple levels of aggregation in data cubes
 - Further reduce the size of data to deal with
- Reference appropriate levels
 - Use the smallest representation which is enough to solve the task
- Queries regarding aggregated information should be answered using data cube, when possible



Automatic Concept Hierarchy Generation

- ❑ Some hierarchies can be automatically generated based on the analysis of the number of distinct values per attribute in the data set
 - ❑ The attribute with the most distinct values is placed at the lowest level of the hierarchy
 - ❑ Exceptions, e.g., weekday, month, quarter, year

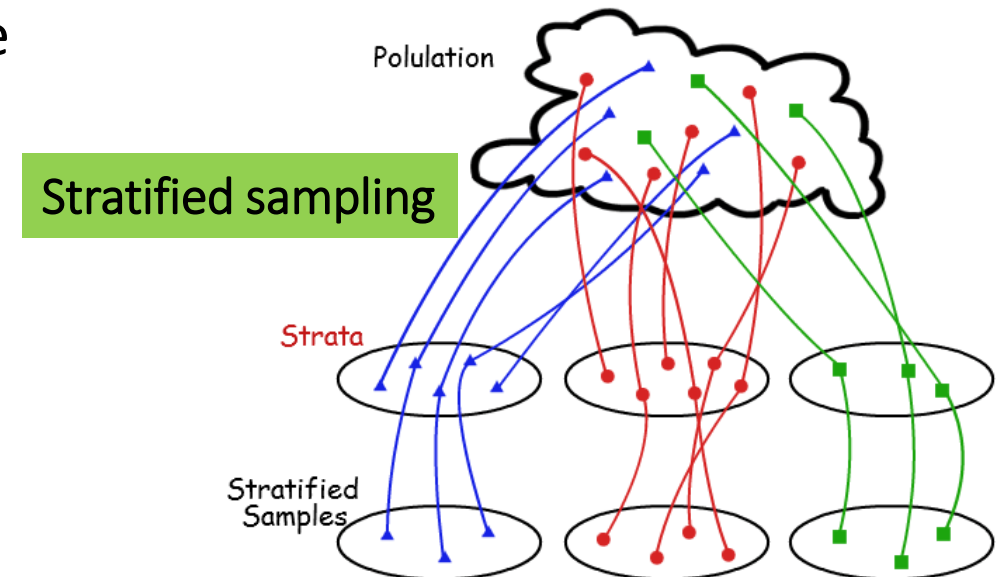
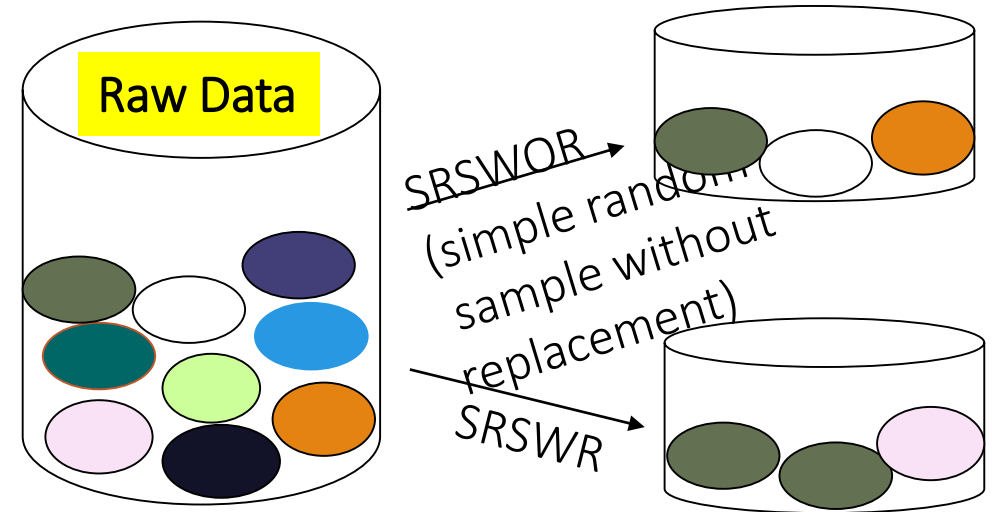


Sampling

- ❑ Sampling: obtaining a small sample s to represent the whole data set N
- ❑ Allow a mining algorithm to run in complexity that is potentially sub-linear to the size of the data
- ❑ Key principle: Choose a **representative** subset of the data
 - ❑ Simple random sampling may have very poor performance in the presence of skew
 - ❑ Develop adaptive sampling methods, e.g., stratified sampling:
- ❑ Note: Sampling may not reduce database I/Os (page at a time)

Types of Sampling

- ❑ **Simple random sampling:** equal probability of selecting any particular item
- ❑ **Sampling without replacement**
 - ❑ Once an object is selected, it is removed from the population
- ❑ **Sampling with replacement**
 - ❑ A selected object is not removed from the population
- ❑ **Stratified sampling**
 - ❑ Partition (or cluster) the data set, and draw samples from each partition (proportionally, i.e., approximately the same percentage of the data)

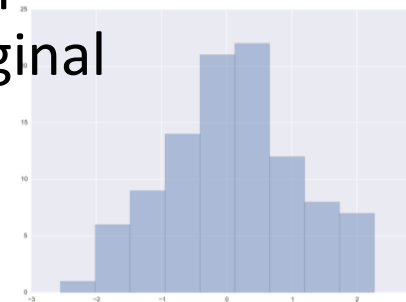
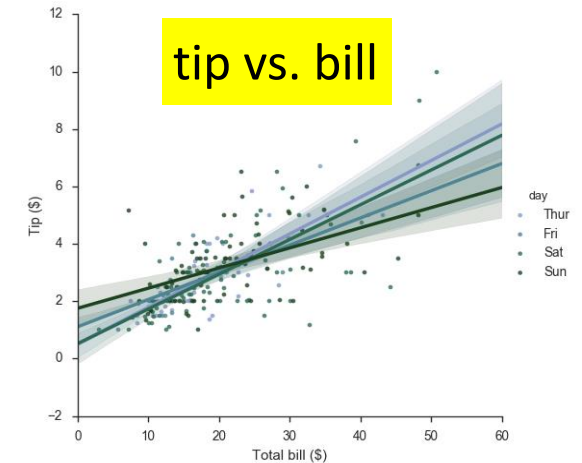


Data Reduction

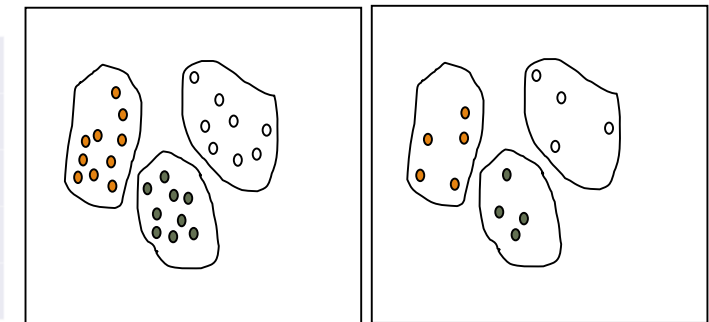
- ❑ **Data reduction:**
 - ❑ Obtain a reduced representation of the data set
 - ❑ much smaller in volume but yet produces *almost* the same analytical results
- ❑ Why data reduction?—A database/data warehouse may store terabytes of data
 - ❑ Complex analysis may take a very long time to run on the complete data set
- ❑ **Methods for data reduction** (also *data size reduction* or *numerosity reduction*)
 - ❑ Regression and Log-Linear Models
 - ❑ Histograms, clustering, sampling
 - ❑ Data cube aggregation
 - ❑ Data compression

Data Reduction: Parametric vs. Non-Parametric Methods

- ❑ Reduce data volume by choosing alternative, *smaller forms* of data representation
- ❑ **Parametric methods** (e.g., regression)
 - ❑ Assume the data fits some model, estimate model parameters, store only the parameters, and discard the data (except possible outliers)
 - ❑ Ex.: Log-linear models—obtain value at a point in m -D space as the product on appropriate marginal subspaces
- ❑ **Non-parametric methods**
 - ❑ Do not assume models
 - ❑ Major families: histograms, clustering, sampling, ...



Histogram

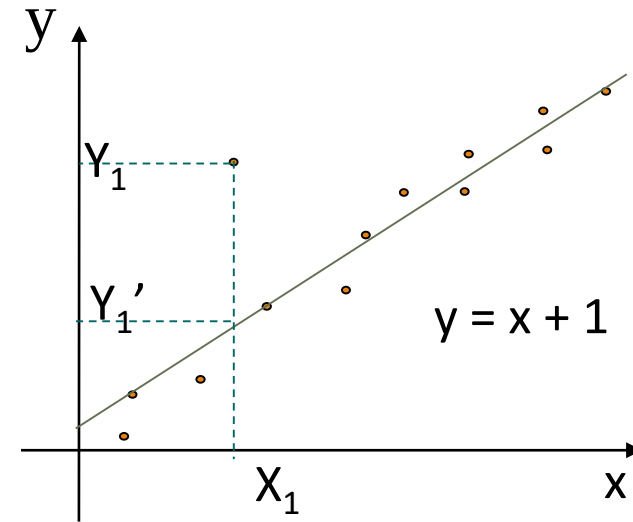


Clustering on the Raw Data

Stratified Sampling

Parametric Data Reduction: Regression Analysis

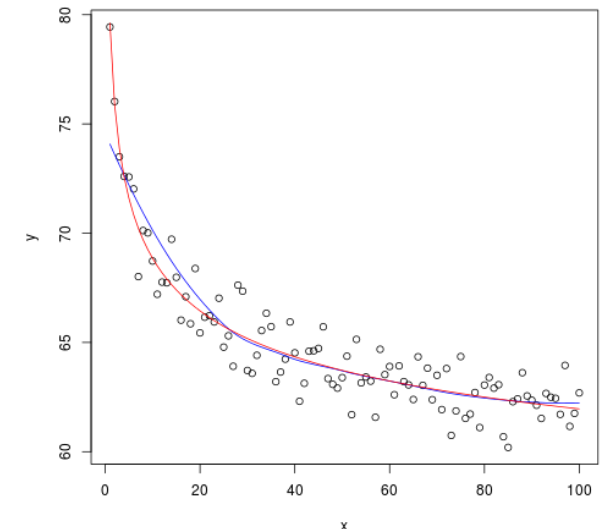
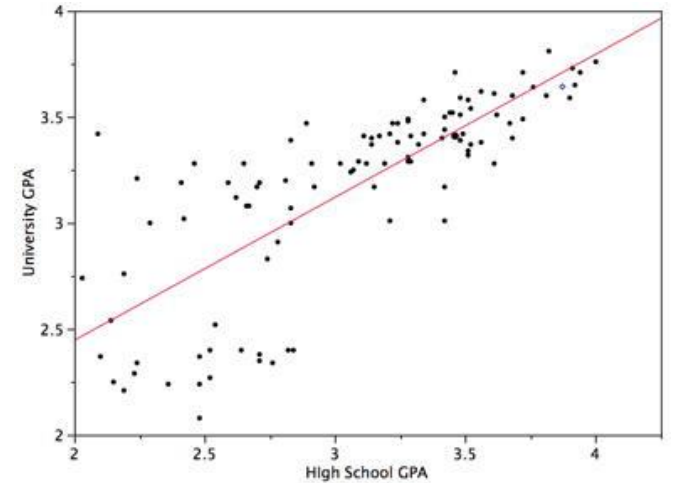
- ❑ Regression analysis: A collective name for techniques for the modeling and analysis of numerical data consisting of values of a **dependent variable** (also called **response variable** or *measurement*) and of one or more **independent variables** (also known as **explanatory variables** or **predictors**)
- ❑ The parameters are estimated so as to give a "**best fit**" of the data
- ❑ Most commonly the best fit is evaluated by using the **least squares method**, but other criteria have also been used



- ❑ Used for prediction (including forecasting of time-series data), inference, hypothesis testing, and modeling of causal relationships

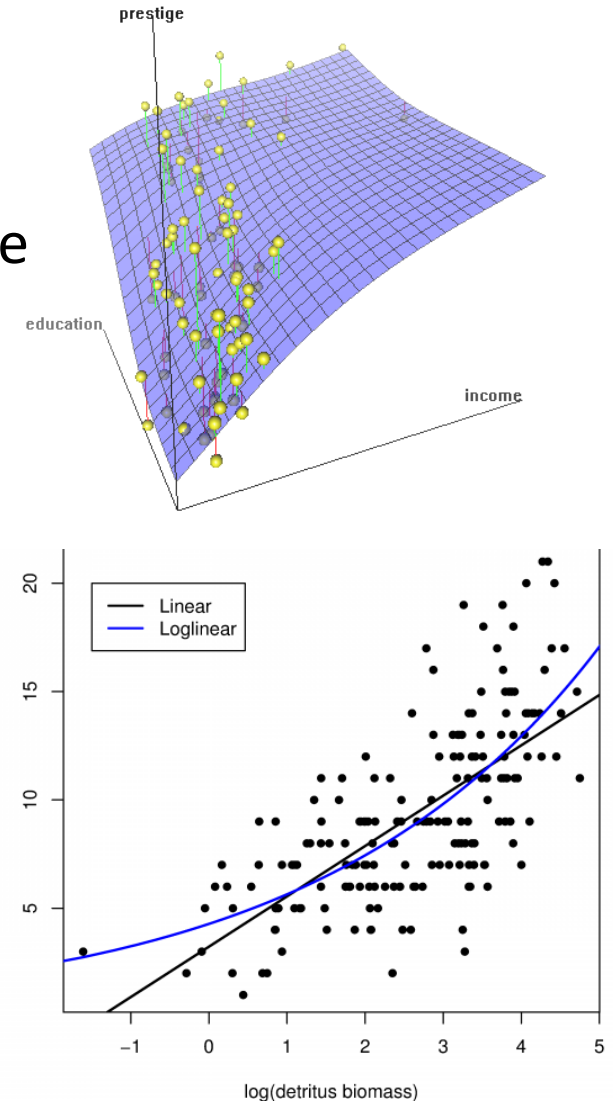
Linear and Multiple Regression

- ❑ Linear regression: $Y = wX + b$
 - ❑ Data modeled to fit a straight line
 - ❑ Often uses the least-square method to fit the line
 - ❑ Two regression coefficients, w and b , specify the line and are to be estimated by using the data at hand
 - ❑ Using the least squares criterion to the known values of $Y_1, Y_2, \dots, X_1, X_2, \dots$
- ❑ Nonlinear regression:
 - ❑ Data are modeled by a function which is a nonlinear combination of the model parameters and depends on one or more independent variables
 - ❑ The data are fitted by a method of successive approximations



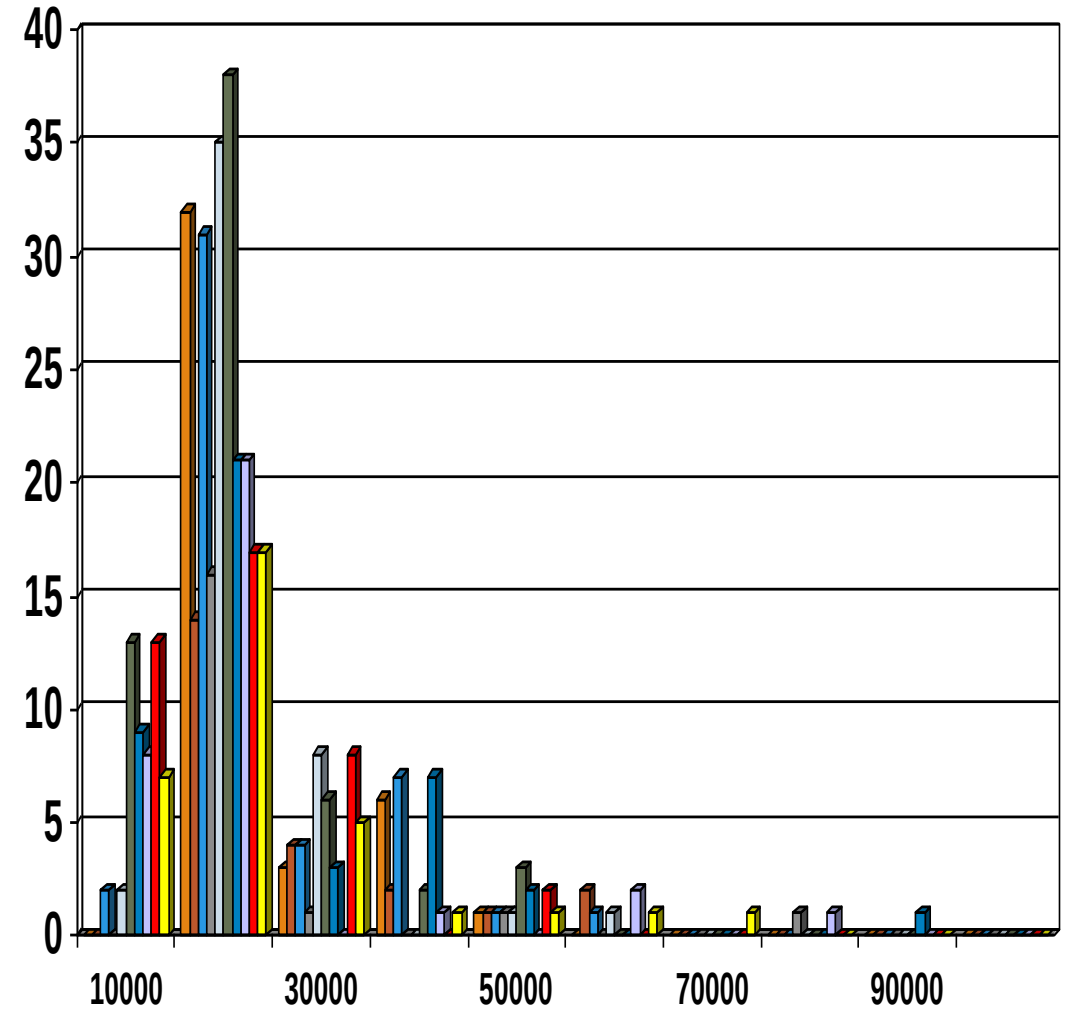
Multiple Regression and Log-Linear Models

- ❑ Multiple regression: $Y = b_0 + b_1 X_1 + b_2 X_2$
 - ❑ Allows a response variable Y to be modeled as a linear function of multidimensional feature vector
 - ❑ Many nonlinear functions can be transformed into the above
- ❑ Log-linear model:
 - ❑ A math model that takes the form of a function whose logarithm is a linear combination of the parameters of the model, which makes it possible to apply (possibly multivariate) linear regression
 - ❑ Estimate the probability of each point (tuple) in a multi-dimen. space for a set of discretized attributes, based on a smaller subset of dimensional combinations
 - ❑ Useful for dimensionality reduction and data smoothing



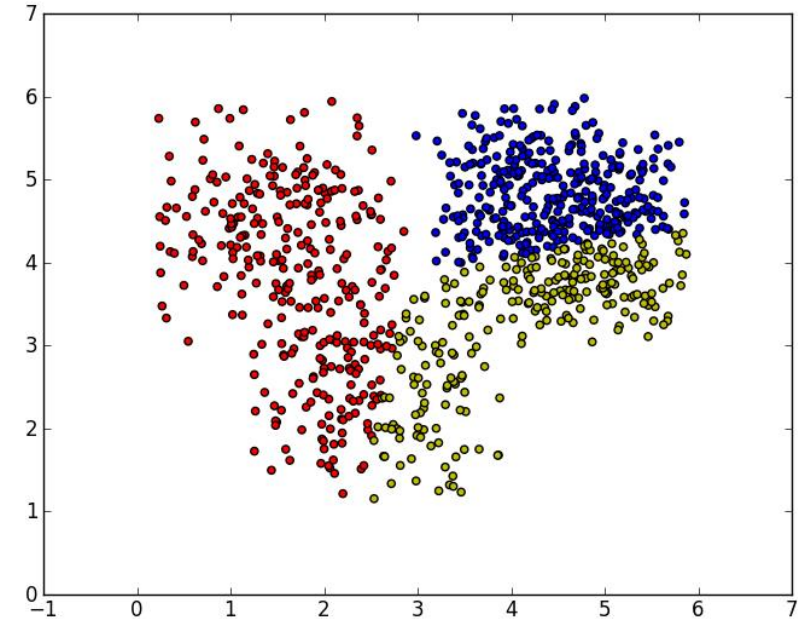
Histogram Analysis

- Divide data into buckets and store average (sum) for each bucket
- Partitioning rules:
 - Equal-width: equal bucket range
 - Equal-frequency (or equal-depth)



Clustering

- ❑ Partition data set into clusters based on similarity, and store cluster representation (e.g., centroid and diameter) only
- ❑ Can be very effective if data is clustered but not if data is “smeared”
- ❑ Can have hierarchical clustering and be stored in multi-dimensional index tree structures
- ❑ There are many choices of clustering definitions and clustering algorithms
- ❑ Cluster analysis will be studied in depth in Chapter 10



Dimensionality Reduction

- ❑ What Is Dimensionality Reduction?
- ❑ Dimensionality Reduction Methods
 - ❑ Principal Component Analysis
 - ❑ Attribute Subset Selection
 - ❑ Nonlinear Dimensionality Reduction Methods

What Is Dimensionality Reduction?

❑ Curse of dimensionality

- ❑ When dimensionality increases, data becomes increasingly sparse
- ❑ Density and distance between points, which is critical to clustering, outlier analysis, becomes less meaningful
- ❑ The possible combinations of subspaces will grow exponentially

❑ Dimensionality reduction

- ❑ Reducing the number of random variables under consideration, via obtaining a set of principal variables

❑ Advantages of dimensionality reduction

- ❑ Avoid the curse of dimensionality
- ❑ Help eliminate irrelevant features and reduce noise
- ❑ Reduce time and space required in data mining
- ❑ Allow easier visualization

Dimensionality Reduction Methods

- ❑ Dimensionality reduction methodologies
 - ❑ **Feature selection:** Find a subset of the original variables (or features, attributes)
 - ❑ **Feature extraction:** Transform the data in the high-dimensional space to a space of fewer dimensions
- ❑ Some typical dimensionality reduction methods
 - ❑ Principal Component Analysis
 - ❑ Attribute Subset Selection
 - ❑ Nonlinear Dimensionality Reduction

Principal Component Analysis – The need

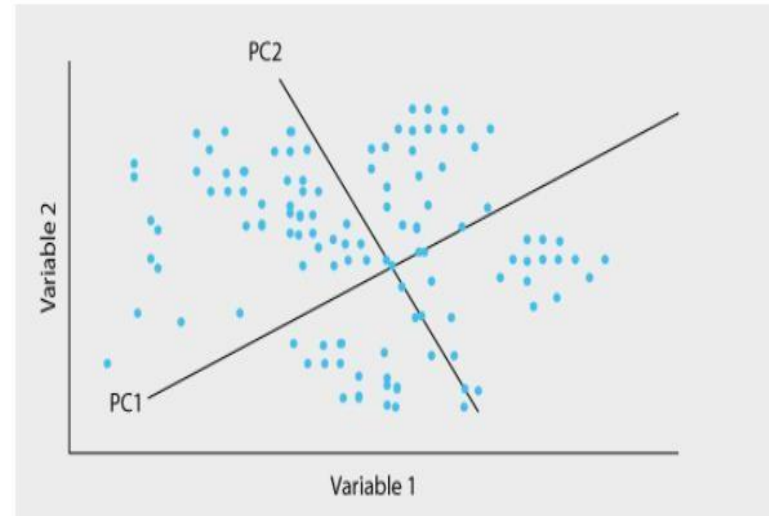
- When the data has too many dimensions, then it becomes a problem for Data Mining.
 - High compute and execution time
 - The risk of compromise in the quality of the model fit.
- When the dimension of data is too high, we need to find a way to reduce it.
- But that reduction has to be done in such a way that we maintain the original pattern of the data.
- PCA – A well known dimensionality reduction technique that maps the data into the space of lower of dimensionality
- Transfer the set of correlated variables into the new set of un-correlated variables
- Form of Un-supervised learning and also used as data pre-processing technique

Principal Component Analysis

- Principal component analysis is a method of extracting important variables known as principal components from a large set of variables available in a data set.
- It represents the original data in terms of its principal components in a new dimension space.
- Principal components (PC) are the directions where there is the most variance, the directions where the data is most spread out.
- There are multiple principal components of a data – each representing the different variance of the data. They are arranged in a chronological order of variance.
- The first PC will capture the most variance i.e. the most information about the data, followed by the second, third and so on.

Principal Component Analysis

- Mathematically, the principal components are the eigenvectors of the covariance matrix of the original dataset.
- Eigenvector corresponds to a direction. Each eigenvector has corresponding eigenvalue
- A eigenvalue is a number that indicates how much variance there is in the data along that eigenvector
- A larger eigenvalue means that that principal component explains a large amount of the variance in the data.

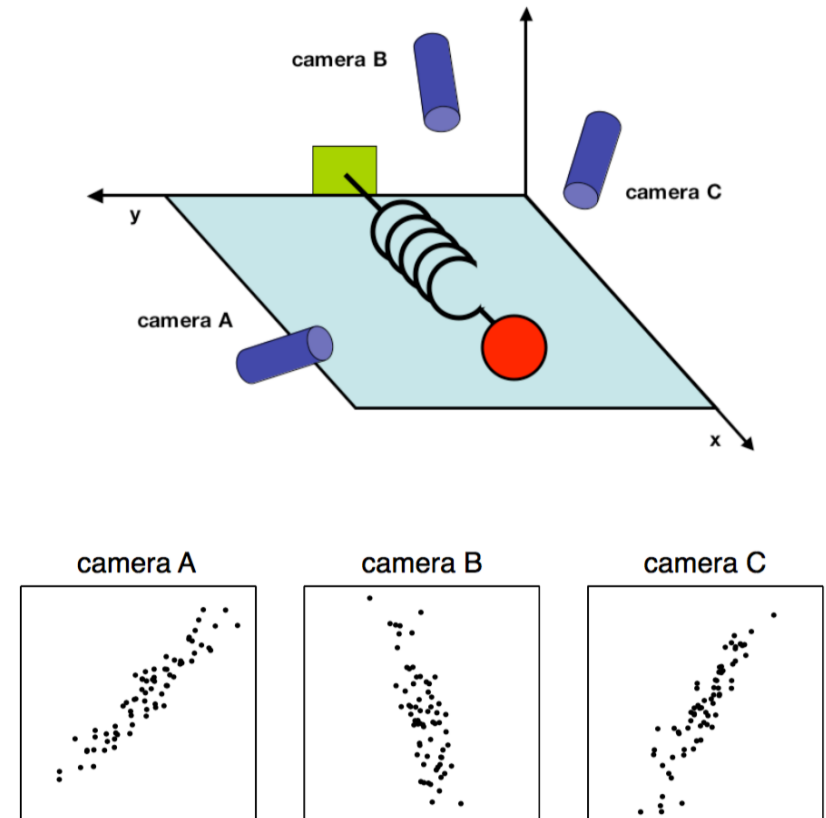


PCA – How it Works ?

- The main purpose of a principal component analysis is the analysis of data to identify and find patterns to reduce the dimensions of the dataset with a minimal loss of information.
- 'N' dimension of data
- Obtain the Covariance matrix
- Calculate 'N' Eigen values and Eigen vectors
- Order the Eigen vectors based on the Eigen values
- The Eigen vector with the highest Eigen value is the principal components of the given data set
- Thus, the given 'N' dimension of data has been reduced into 'p' principal components finally.

Principal Component Analysis (PCA)

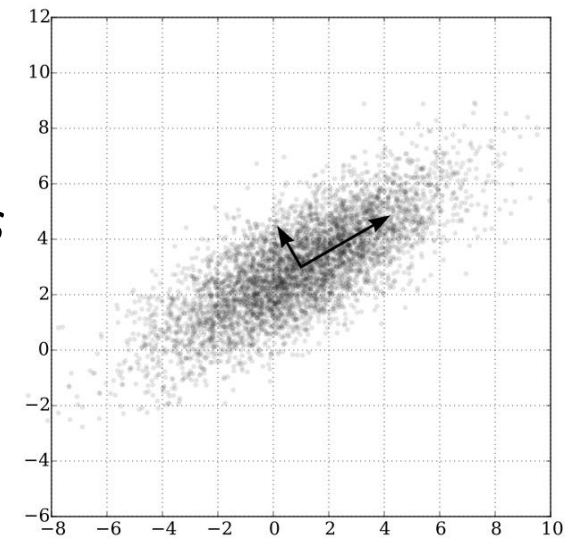
- ❑ PCA: A statistical procedure that uses an orthogonal transformation to convert a set of observations of possibly correlated variables into a set of values of linearly uncorrelated variables called *principal components*
- ❑ The original data are projected onto a much smaller space, resulting in dimensionality reduction
- ❑ Method: Find the eigenvectors of the covariance matrix, and these eigenvectors define the new space



Ball travels in a straight line. Data from three cameras contain much redundancy

Principal Component Analysis (Method)

- Given N data vectors from n -dimensions, find $k \leq n$ orthogonal vectors (*principal components*) best used to represent data
 - Normalize input data: Each attribute falls within the same range
 - Compute k orthonormal (unit) vectors, i.e., *principal components*
 - Each input data (vector) is a linear combination of the k principal component vectors
 - The principal components are sorted in order of decreasing “significance” or strength
 - Since the components are sorted, the size of the data can be reduced by eliminating the *weak components*, i.e., those with low variance (i.e., using the strongest principal components, to reconstruct a good approximation of the original data)
- Works for numeric data only



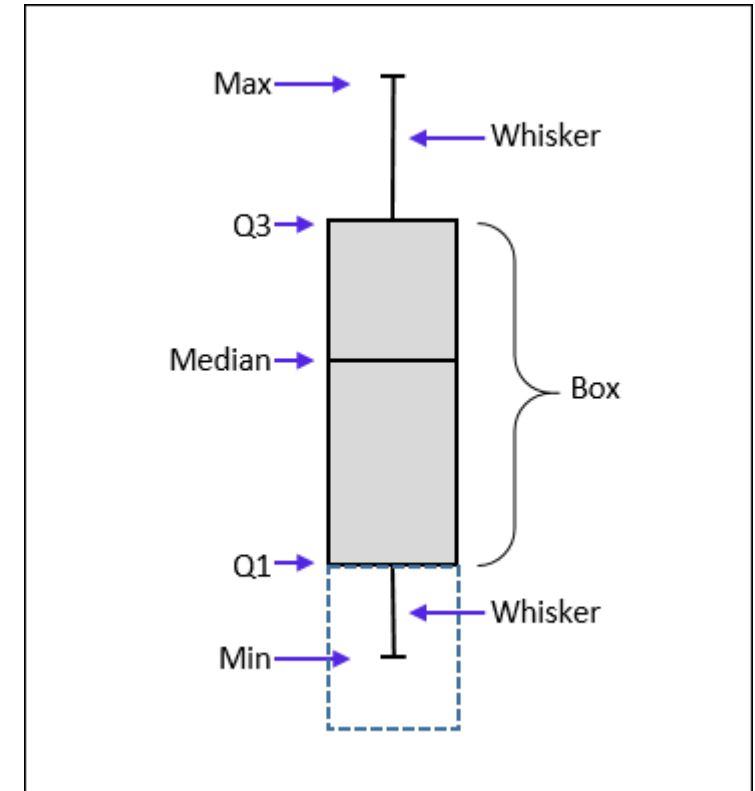
Ack. Wikipedia: Principal Component Analysis

Graphic Displays of Basic Statistical Descriptions

- ❑ **Boxplot:** graphic display of five-number summary
- ❑ **Histogram:** x-axis are values, y-axis repres. frequencies
- ❑ **Quantile plot:** each value x_i is paired with f_i indicating that approximately $100 f_i \%$ of data are $\leq x_i$
- ❑ **Quantile-quantile (q-q) plot:** graphs the quantiles of one univariant distribution against the corresponding quantiles of another
- ❑ **Scatter plot:** each pair of values is a pair of coordinates and plotted as points in the plane

Measuring the Dispersion of Data: Quartiles & Boxplots

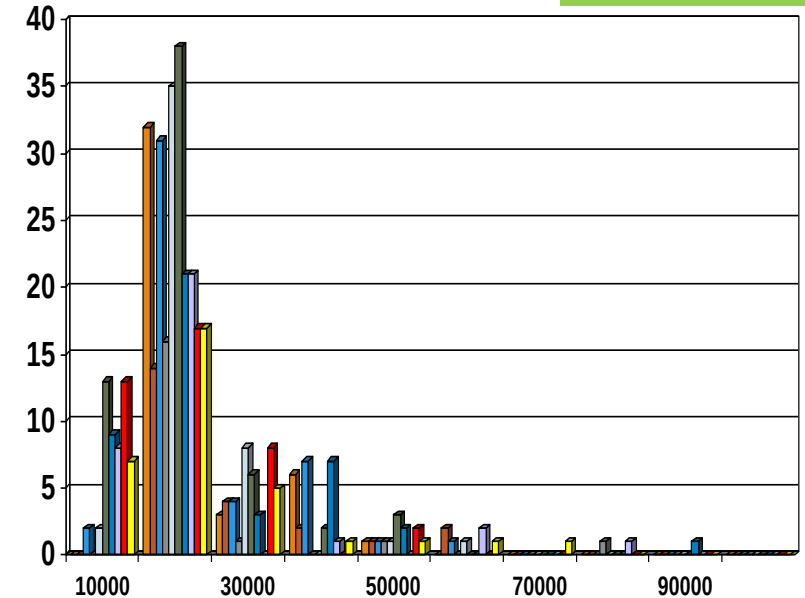
- ❑ **Quartiles:** Q_1 (25th percentile), Q_3 (75th percentile)
- ❑ **Inter-quartile range:** $IQR = Q_3 - Q_1$
- ❑ **Five number summary:** min, Q_1 , median, Q_3 , max
- ❑ **Boxplot:** Data is represented with a box
 - ❑ Q_1 , Q_3 , IQR: The ends of the box are at the first and third quartiles, i.e., the height of the box is IQR
 - ❑ Median (Q_2) is marked by a line within the box
 - ❑ Whiskers: two lines outside the box extended to Minimum and Maximum
 - ❑ Outliers: points beyond a specified outlier threshold, plotted individually
 - ❑ **Outlier:** usually, a value higher/lower than $1.5 \times IQR$



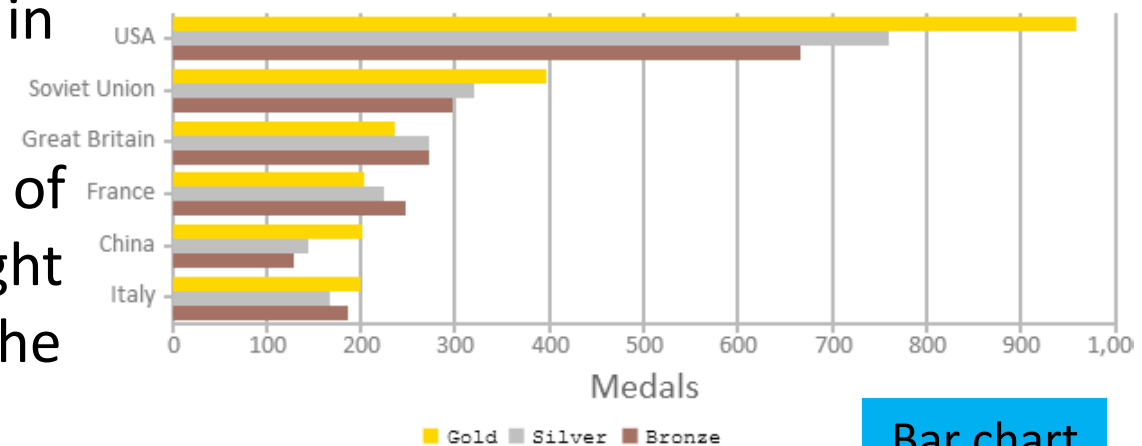
Histogram Analysis

- ❑ Histogram: Graph display of tabulated frequencies, shown as bars
- ❑ Differences between histograms and bar charts
 - ❑ Histograms are used to show distributions of variables while bar charts are used to compare variables
 - ❑ Histograms plot binned quantitative data while bar charts plot categorical data
 - ❑ Bars can be reordered in bar charts but not in histograms
 - ❑ Differs from a bar chart in that it is the area of the bar that denotes the value, not the height as in bar charts, a crucial distinction when the categories are not of uniform width

Histogram

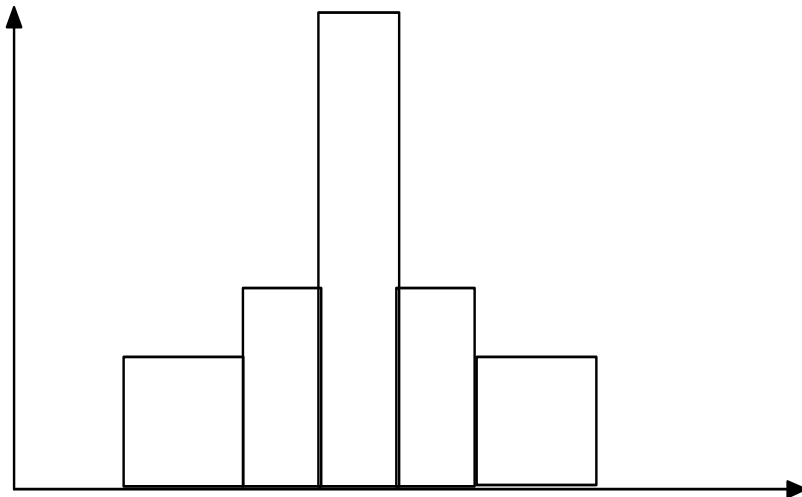
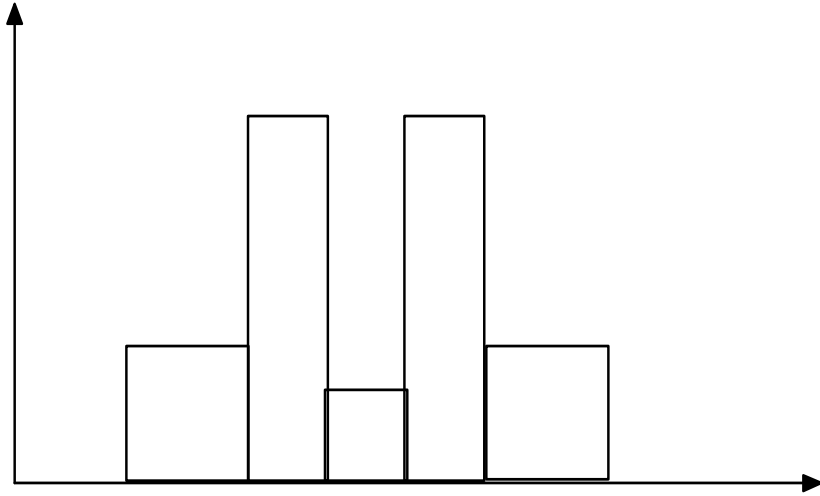


Olympic Medals of all Times (till 2012 Olympics)



Bar chart

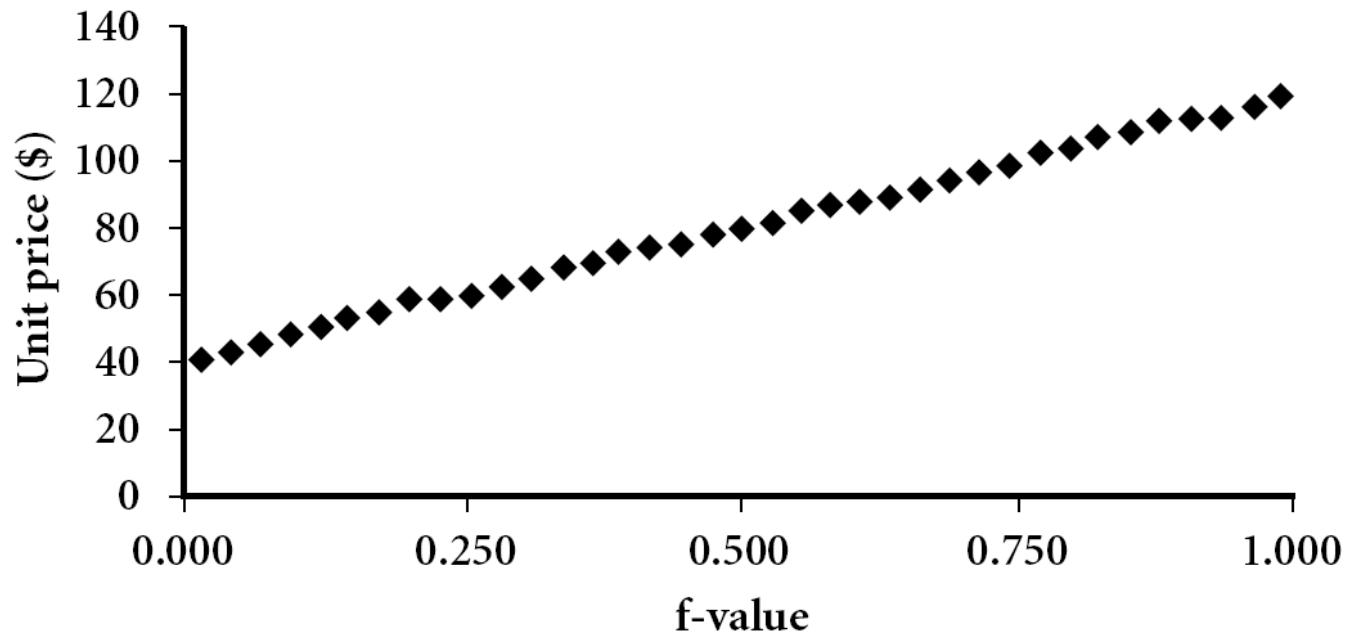
Histograms Often Tell More than Boxplots



- The two histograms shown in the left may have the same boxplot representation
- The same values for: min, Q1, median, Q3, max
- But they have rather different data distributions

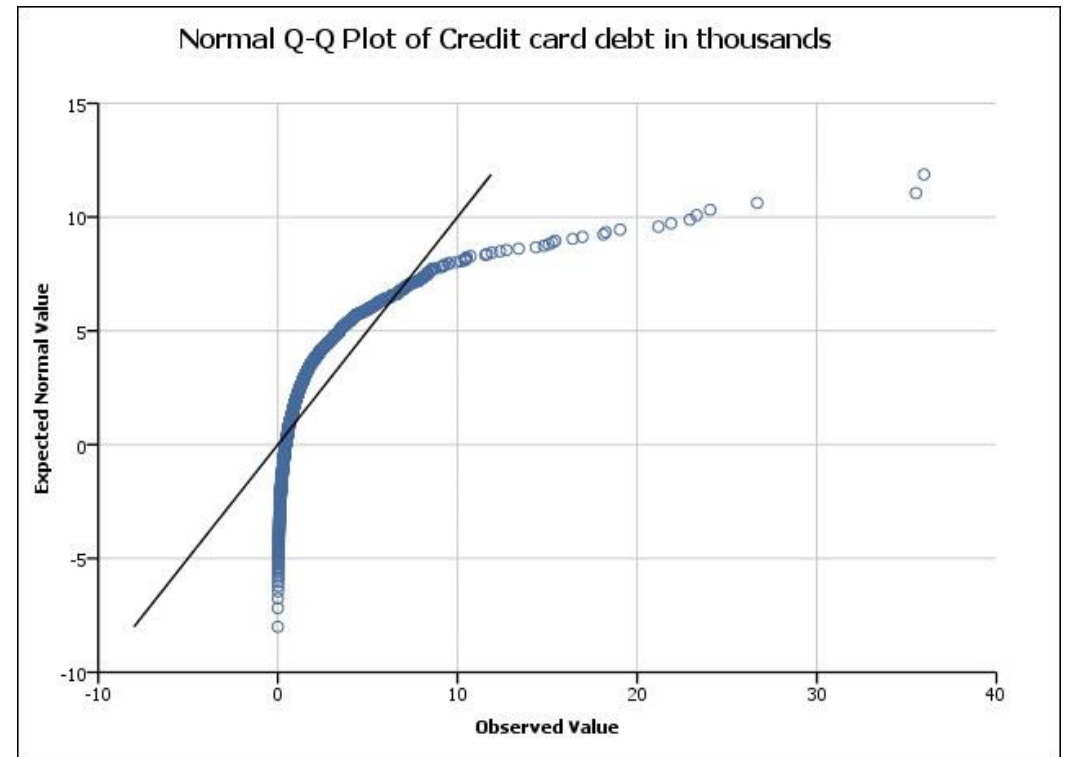
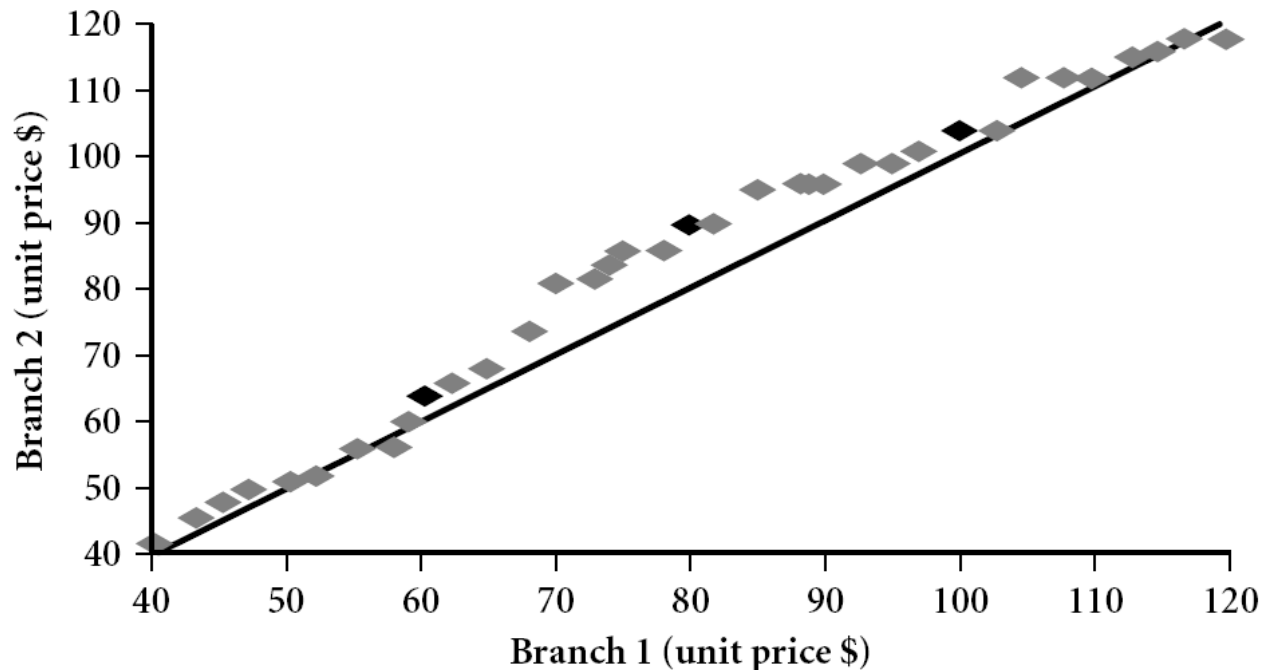
Quantile Plot

- ❑ Displays all of the data (allowing the user to assess both the overall behavior and unusual occurrences)
- ❑ Plots **quantile** information
 - ❑ For a data x_i data sorted in increasing order, f_i indicates that approximately 100 f_i % of the data are below or equal to the value x_i



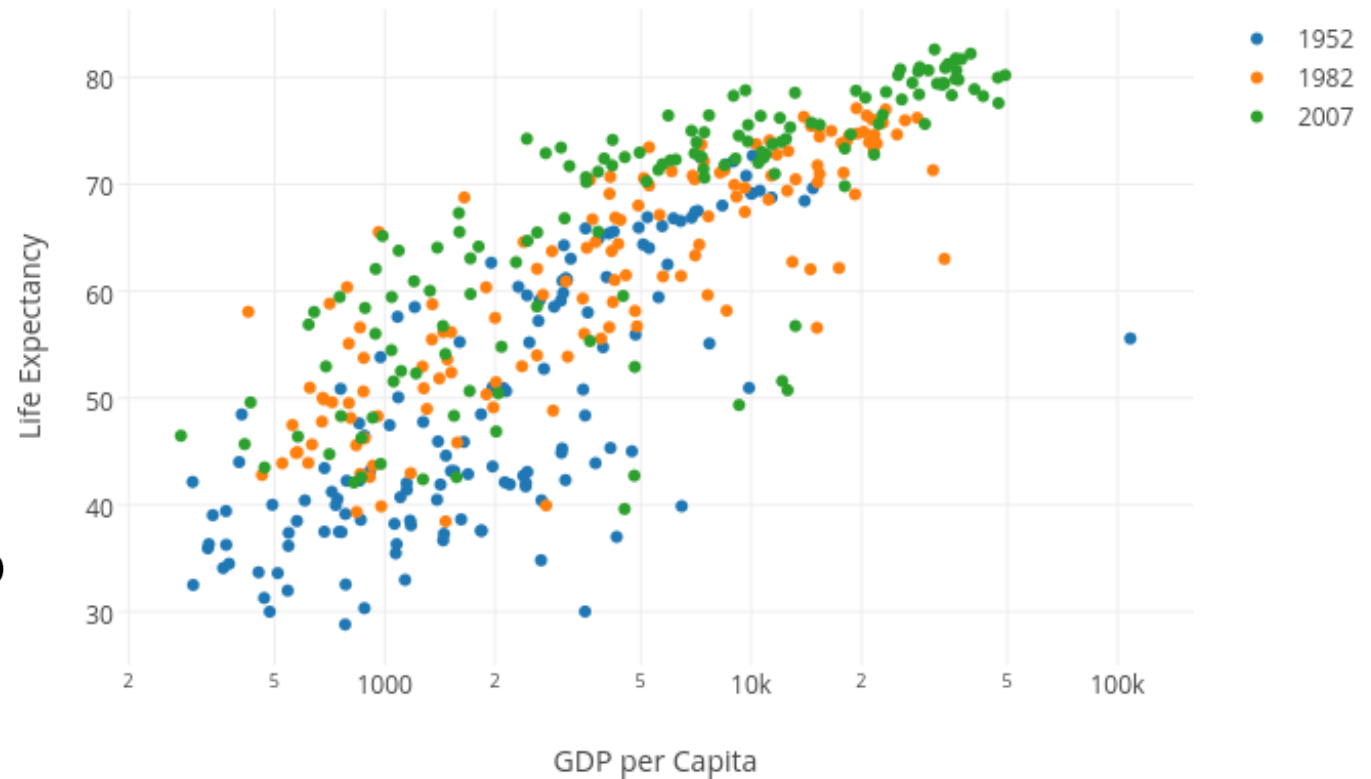
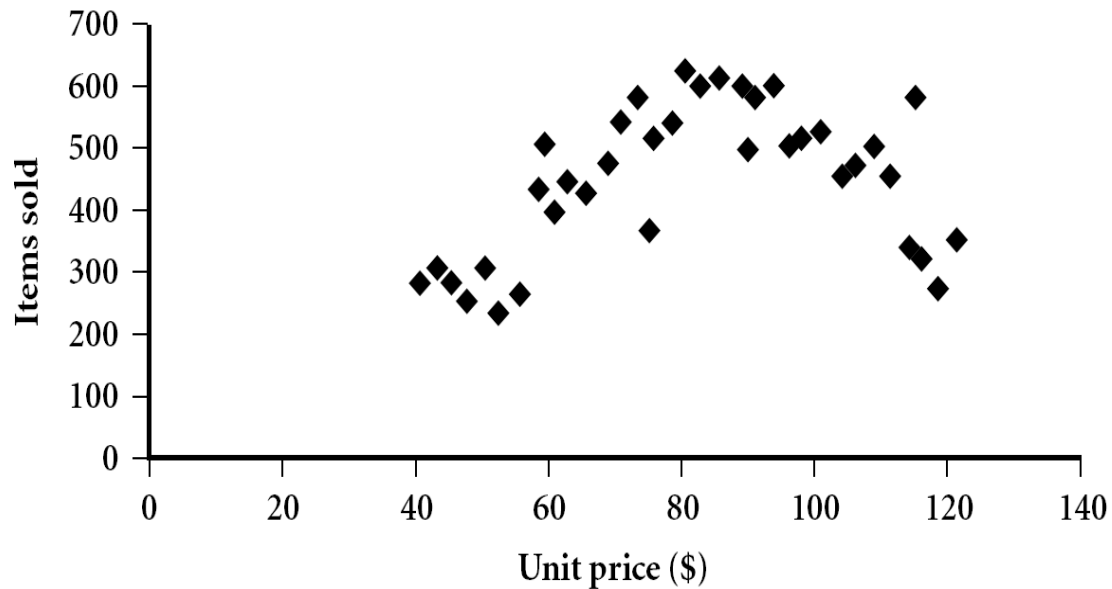
Quantile-Quantile (Q-Q) Plot

- ❑ Graphs the quantiles of one univariate distribution against the corresponding quantiles of another
- ❑ View: Is there is a shift in going from one distribution to another?
- ❑ Example shows unit price of items sold at Branch 1 vs. Branch 2 for each quantile. Unit prices of items sold at Branch 1 tend to be lower than those at Branch 2

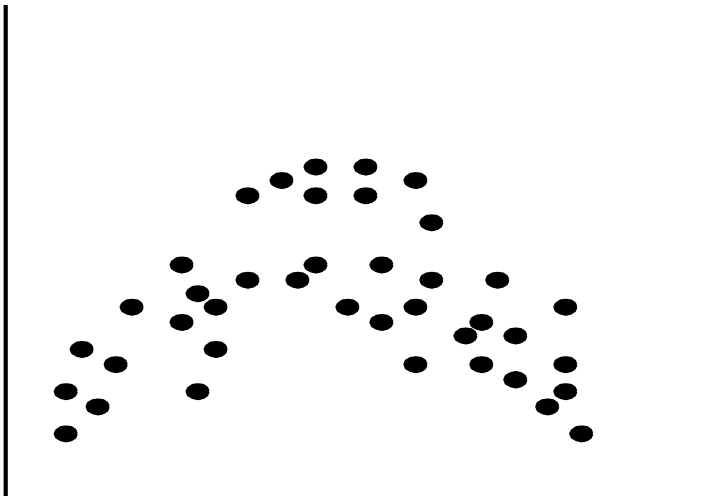
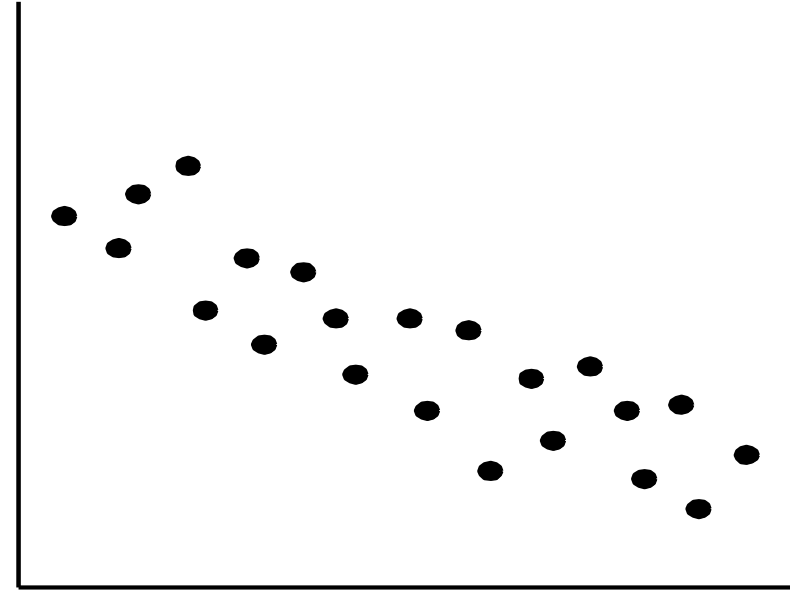
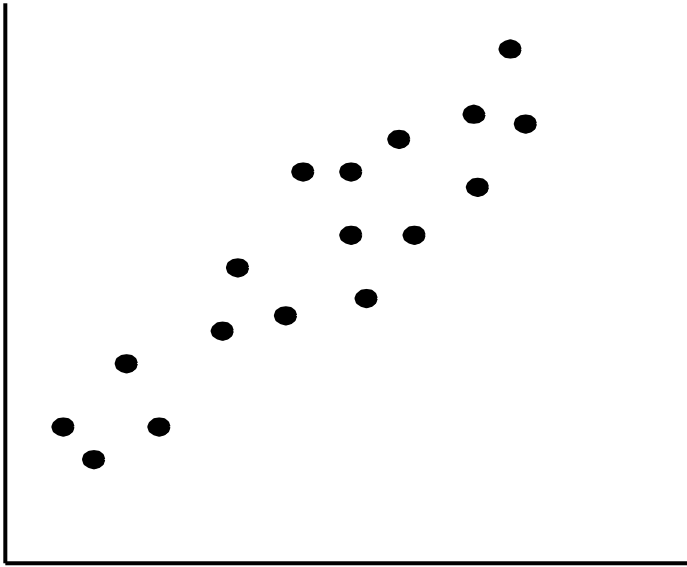


Scatter plot

- ❑ Provides a first look at bivariate data to see clusters of points, outliers, etc.
- ❑ Each pair of values is treated as a pair of coordinates and plotted as points in the plane



Positively and Negatively Correlated Data



- ❑ The left half fragment is positively correlated
- ❑ The right half is negative correlated

Uncorrelated Data

